ASSESSMENT BRIEF

Coursework 1

Programme	BSc Computing		
Module Title	Data Science		
Module Code	CMP-N205-0		
Module Level	6	**Assessment Type(s)**	Report
Word Length / Duration	1000 words	**% contribution to module mark**	30%
Deadline (date & time) for Submission	17th April 2026 (Friday)	**Format/Location of submission**	Moodle
Assessment Feedback date:	24th April 2026 (Friday)		

Learning Outcomes

This assessment has been designed to provide you with an opportunity to demonstrate your achievement of the learning outcomes listed below.

By successfully completing this assessment, you will be able to

- Collecting and Assessing Quality of Data that will be used.
- Use and apply appropriate models for analysis.
- Derive insights from the analysis results.
- Investigate and discuss potential issues on the business-related challenges.
- Compare and contrast different uses of Machine Learning concepts.

Employability and Professional Skills

This assessment has been designed to provide you with an opportunity to demonstrate your achievement of the following skill(s) which is(are) critical in professional context and jobs.

By successfully completing this assessment, you will be able to

- Collecting and Assessing Quality of Data that will be used.
- Use and apply appropriate models for analysis.
- Derive insights from the analysis results.
- Investigate and discuss potential issues on the business-related challenges.

	• Compare and contrast different uses of Machine Learning concepts.

Assessment Requirements

In practice, the skills of Data Scientists are often required to identify problems that need solving in a company, institution, organisation or in an industry. Then the solution is found out by analysing the data that is present/available for that area of application.

Typically, the first step of building that solution will involve formulating **a hypothesis** that may consist of one or more **Data Science** problemsidentified from the data, e.g., if the data you will use contains carbon emission rate (numeric data) because of the transportation (different transportations such as bus, taxi and private car) then you maybuild a hypothesis like "analysing the data will contribute to the carbon reduction plan government has to execute before 2030", which needs to be supported by appropriate questions/problems that relevant to it. These problems are often related to some business/ operational/ management/economic decisions that the authority that is carrying out the analysis, need to be solved.

In coursework 1, you will ***take the role of a Data Scientist and resolve the business/ operational/ management/ economic problems*** of a fictitious/real company, institution, organisation or for an industry. For this, you must first collect a Dataset.

Criteria of Collected Dataset

- Number of observations / sample size: at least 500
- Number of attributes / variables: at least 5
- At least 10% datapoints need to be cleaned (i.e. clear formatting; irrelevant data; duplicates; missing data; outliner; data type...)

You are advised to collect the Dataset from online sources, or Data Banks which are opensource and available online. Then, you must brainstorm and identify some business/ operational/ management problems that can be solved using your chosen dataset, this will form your hypothesis.

Finally, you will ***propose to solve those defined problems usingone of the methods you have learned in this module*** e.g., Naïve Bayes, KNN, Regression, Clustering, Association Rule Mining etc.

Note, that, you will NOT HAVE TO use the whole dataset or, exactly the way you collect it. You can usedifferent preprocessing steps on your dataset and analyse only the data you would extract from the whole dataset. The technique you will need to use for analysing your data and finding the solutions toyour defined problems will, therefore, largely depend on your pre-processed data.

There are numerous numbers of datasets available online. For your ease, here some of the online sources and Databank links are provided. Note that, you can also go for other sources

which aren't mentioned here:

1. https://data.gov.uk/
2. https://ukdataservice.ac.uk/help/other-data-providers/open-data-resources/
3. https://data.london.gov.uk/
4. https://www.kaggle.com/datasets

Marking Criteria

1. Providing the source of the Data correctly, providing the source of the data correctly done with appropriate Discussion (15%)
2. Pre-processing of Data Any pre-processing encountered is discussed (20%)
3. Interpreting the Data correctly for the purpose - The attributes to be used for the proposed solution are clearly presented (20%)
4. Building a hypothesis - The problems to be solved that are derived from the data. (20%)
5. Choosing appropriate method/ technique for the purpose - Selected method/technique is appropriate for the kind of Data (20%)
6. Report Submission - ability to provide a submission that meets the requirements given. (5%)

Assessment Success Guidance

1. A ***discussion about collected dataset*** (within 400-500 words):

2. A ***discussion about the technique*** you will use for your proposed solution (within 300-400 words):

Assessment Guidance Support and Formative Feedback

All the following must be submitted within the due date to be considered for a complete submission for this assessment.

(1) Report with both the following section 1 and section 2

(2) Python interactive notebook

(3) Raw dataset in Excel format

(4) Cleaned dataset in Excel format

in order to be considered for a complete submission for this part.

Your submission must be a single report (.PDF) and should contain:

Section 1 A *discussion about collected dataset* (within 400-500 words):

a) Discussion about the source of the Dataset along with the URL of the source from where the dataset has been collected/downloaded. Remember, providing the sourceof the dataset is vital.

b) Discussion on the hypothesis that you will build from your data. This needs clear, precise problem statements that must have some value in business/operational/management/economic terms. In order to demonstrate your skill, you will need to have at least 3 such problems in your hypothesis. You will also need to explain why these problems are important for your fictitious/real authority for whom you are building these solutions.

c) Discussion about the preprocessing steps that you have utilized for making the data appropriately usable for your purpose (You can use Microsoft Excel for preprocessingthe Data, see Week-3 contents). Please note, you will need to provide screenshots forthe pre-processed data along with the screenshot of the raw/collected dataset in thissection.

Section 2 A *discussion about the technique* you will use for your proposed solution (within 300-400 words):

d) Your proposed solution can be achieved by utilizing Supervised/ Unsupervised Learning technique that you have learnt from week-5 to week10 (KNN, Linear Regression, Naïve Bayes, K-Means Clustering, Association Rule Mining etc.). ****You just choose one technique appropriately from the Enlisted ones for the Coursework.*

e) You must provide your argument(s)/reason(s) appropriately on your choice of method/technique and how it is appropriate for the kind of data (As attributes) you have chosen and processed for this task.

****Note that, **<u>you are just to propose the kind of technique</u>** you will use with proper argument and the actual implementation goes in coursework 2.*

Additional Information (if required):

Assessment Criteria (Grade Boundaries or Rubric):

Criteria	Excellent	Satisfactory	Not Satisfactory	Not Attempted
Providing the source ofthe Data correctly, Providing the source ofthe data correctly donewith appropriate Discussion **(15%)**	The source of data is provided is correct and the discussion is appropriate	The source of data is provided is correct and the discussion is somewhat appropriate	The source of data is provided is correct, but no appropriate discussion provided.	The source of data is not provided, and discussion reflected
Pre-processing of Data Any pre-processing encountered is discussed **(20%)**	Pre-processing steps followed are presented correctly with appropriate discussion.	Pre-processing steps followed are presented somewhat correctly with some discussion.	Pre-processing steps followed are discussed to some extent but is not clear.	No clue about any pre-processing step is provided.
Interpreting the Datacorrectly for the purpose The attributes to be used for the proposedsolution are clearly presented **(20%)**	The discussion clearly reflects the Interpretation of the extracted and/or, pre-processed Data (attributes)	The discussion is not very clear but still reflects the Interpretation of the extracted and/or, pre-processed Data somewhat correctly	The discussion provided is not clear but somehow the interpretation got reflected	No reflection about the interpretation of the Data reflected
Building a hypothesis The problems to be solved that are derivedfrom the data. **(20%)**	The hypothesis is clear, precise with properly defined and justified set of at least 3 problems.	The hypothesis is not precise, with at least 3 problems that are defined but their need is not properly justified.	The hypothesis is lacking with less than 3 problems given that may/may not be adequately justified.	No hypothesis was developed, and no problems were identified.
Choosing appropriate method/ technique forthe purpose Selected method/technique is appropriate for the kind	Selected Method/technique is appropriate and properly justified.	Selected Method/technique is somewhat appropriate, and/or the justification is okay but is lacking.	Selected Method/technique is not appropriate and/or the justification is severely lacking.	No evidence of choosing a method is there and no justification is given.

of Data **(20%)**				
Report Submission ability to provide a submission that meets the requirements given. **(5%)**	Correct files submitted only with no issues	Correct files submitted only but some issues with files provided	Correct files submitted but with unwanted files and possibly issues in files provided	Incorrect/no files were submitted.

Contact for Queries/ who you can contact for further information or queries

Should you have any question, please do not hesitate to contact Dr Michael Ng at michael.ng@roehampton.ac.uk or Dr Yusuf Izmirlioglu at Yusuf.Izmirlioglu@roehampton.ac.uk

Ethical Requirements

Academic Misconduct:

"Academic integrity and honesty are fundamental to the academic work you produce at the University of Roehampton. You are expected to complete coursework which is your own and which is referenced appropriately. The university has in place measures to detect academic dishonesty in all its forms. If you are found to be cheating or attempting to gain an unfair advantage over other students in any way, this is considered academic misconduct, and you will be penalised accordingly."

Further details about "Student Code of Conduct" and "Disciplinary Regulations" can be found at:

https://www.roehampton.ac.uk/corporate-information/policies/

Use of Artificial Intelligence (AI)

The assessment is designed so that the use of AI during the assessment is irrelevant or impossible. You are welcome to use AI to support learning and preparation for this assessment. Please make sure you read and understand the assessment guidelines and ask your Module Leader if you have any questions. You can find the guidance on the use of AI here*.

https://roehamptonprod.sharepoint.com/sites/portal/information/LTEU/Documents/4b%20Student%20Guidelines%20on%20the%20use%20of%20AI_.docx

**You can find the student guidance on the use of AI in the Library and Nest.*

Referencing

All required references are available on Moodle.

Mitigating circumstances/late penalties

Sometimes circumstances outside of your control may affect your studies and might prevent you from submitting work on time or attending an exam.

The University offers the ability for students to request additional time to complete an assessment or to defer an examination to a later date. If you are finding yourself in such a situation, please speak to your Academic Guidance Tutor, the Roehampton Student Union (RSU) or someone in the **Wellbeing team** first, who can support you. Further details can be found on the mitigating circumstances portal.

If you do not apply for or are not approved for Mitigating Circumstances, late penalties will apply. If work is submitted up to 14 days late, the mark will be capped at 50%; if it is over 14 days late, it will not be marked.

Resubmissions and Reassessment

If you are required to resubmit this assessment or take part in reassessment, you will be notified via Moodle and your student email. Please ensure you check both regularly. Any reassessment tasks will follow the same learning outcomes and criteria.

Submission Checklist

Before you submit, ask yourself:

Have I fully answered the assessment brief?

Have I met the word count and formatting requirements?

Is my referencing complete and accurate?

Have I declared any AI use honestly?

Have I proofread my work?

Am I submitting through the correct platform before the deadline?